Investment Demand Shock and Business Cycle

A Bayesian estimation of a DSGE model

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Motivation

- Investment is the most volatile component of the business cycle; a long literature treats **investment shocks** as a main driving force.
- **Justiniano, Primiceri & Tambalotti (2010):** an estimated DSGE attributes a large share of U.S. fluctuations to a *supply-side* investment-specific technology (IST) shock.
- **The comovement puzzle:** supply-side IST shocks struggle to reproduce the joint positive comovement of consumption, investment, and inflation seen in the data.
- **Nirei (2015):** an *investment demand* (ID) shock—built from firm heterogeneity and lumpy investment—*can* generate this comovement, but lacks empirical evidence.

This paper

Estimate a DSGE model with **both** an IST shock and an investment demand shock to quantify how important the investment demand shock is for the U.S. business cycle.

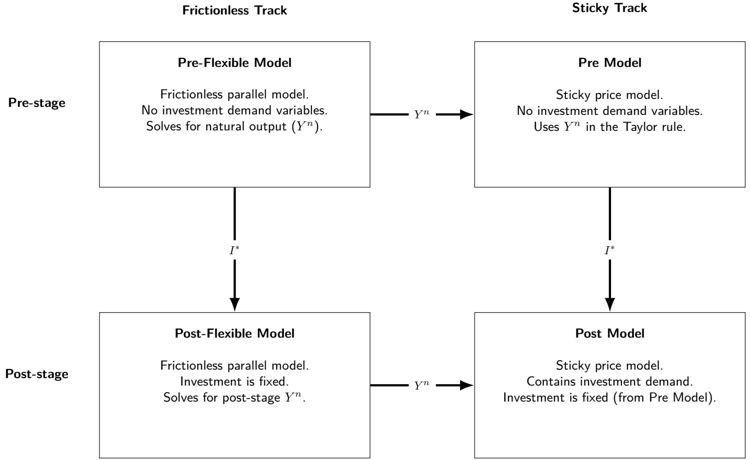
Specifying the investment demand shock

- The micro origin (idiosyncratic productivity + lumpy investment) appears *at the aggregate level* as a deviation of investment from its shock-free decision ⇒ a reduced-form shock $\mu_{d,t}$:

$$I_t = \mu_{d,t}\, I_t^*, \qquad \ln \mu_{d,t} = (1-\rho_d)\ln \mu_d + \rho_d \ln \mu_{d,t-1} + \varepsilon_{d,t}.$$

- Households would otherwise undo the shock, so the equilibrium is solved in **two steps**:
    1. solve the baseline model (no ID shock) for the endogenous investment I_t^*;
    2. reveal $\mu_{d,t}$, *fix* $I_t = \mu_{d,t} I_t^*$, and re-solve the equilibrium for consumption, labor, bonds and utilization.
- Natural output for the Taylor rule is computed from **frictionless parallel models** in each stage (next slide).

The two-step solution and parallel models



Pre-stage solves the endogenous investment I^* and natural output Y^n; the post-stage *fixes* $I = \mu_d I^*$ and re-solves. Frictionless tracks ($\theta_w = 0$) supply Y^n for the Taylor rule.

A toy model of idiosyncratic shocks and lumpy investment

Why is it legitimate to use a reduced-form investment demand shock?

Standard New-Keynesian CES aggregator and demand:

$$Y_t = \Big(\int_0^1 y_{jt}^{\frac{\eta-1}{\eta}} \, dj \Big)^{\frac{\eta}{\eta-1}}, \qquad y_{jt} = \Big(\frac{p_{jt}}{P_t} \Big)^{-\eta} Y_t, \qquad y_{jt} = a_{jt} k_{jt}.$$

Linear-in-capital technology $y_{jt} = a_{jt} k_{jt}$. Combining demand and production, real revenue is

$$r_{jt}(a_{jt}, k_{jt}) = \frac{p_{jt} y_{jt}}{P_t} = (a_{jt} k_{jt})^{1-\frac{1}{\eta}} \, Y_t^{\frac{1}{\eta}}.$$

- Concave in own capital ($\eta > 1$): diminishing returns through the demand curve.
- Revenue, hence the desired capital, scales with aggregate output Y_t — the channel through which one firm's action will later spill over to all firms.

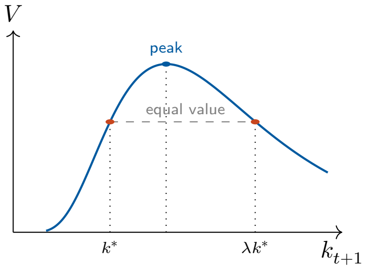
Lumpy investment and the investment threshold

Firms solve

$$\max_{\{x_{t+k}\}} \ \mathbb{E}_t \sum_{k\geq 0} \Lambda_{t,t+k}(r_{j,t+k} - x_{j,t+k}),$$

$$\text{s.t. } k_{j,t+1} = (1-\delta)k_{j,t} + x_{j,t},$$

$$x_{j,t+k} \in \{0, \ \lambda k_{j,t+k}\}.$$



Indifference at k^* (the firm knows its next-period technology) pins down the threshold:

$$k^*_{j,t+1} = a^{\eta-1}_{j,t+1}\,\Phi_t(\Lambda_t, \Lambda_{t+1})\,Y_{t+1}$$

- Desired capital scales with aggregate output Y_{t+1} — the spillover channel that will carry one firm's action to all firms.

Normalize a firm's position by its **distance to the threshold**

$$s_{jt} = \frac{\log(k_{jt}/k^*_{jt})}{\log \lambda},$$

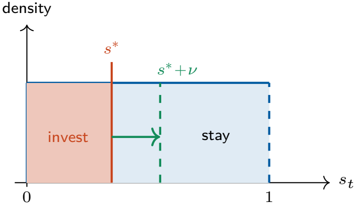
so that investing means jumping up exactly one notch ($s \mapsto s+1$) in these units.

Investing fraction and idiosyncratic shock

- Assume $a_t \in \{a_L, a_H\}$ and $s_t \mid a_t \sim U[0,1)$ under stationary distribution.
- Macro variables are constant: $Y = Y_t = Y_{t+1}$, $\Phi = \Phi_t = \Phi_{t+1}$

Without investing, under stationary distribution position drifts by

$$s_{j,t+1} = s_{j,t}(a) - \underbrace{\frac{-\log(1-\delta) + (\eta-1)\log\frac{a'}{a}}{\log\lambda}}_{s^*(a,a')}.$$



- Since $s_t \sim U[0,1)$, $s^*(a,a')$ is the **investing fraction**
- Add an i.i.d. shock ε on a' *only* in group $(a,a') = (b,b')$:

$$s^*(b,\varepsilon b') = s^*(b,b') + \underbrace{\frac{(\eta-1)\log\varepsilon}{\log\lambda}}_{\nu}.$$

- The cutoff in that group rises by ν, so an extra ν-mass invests.

Propagation through the threshold

More investment in (b, b') raises *future output* of this group, which is a part of *future aggregate output*:

$a = H$	$\omega(H, L)$	$\omega(b, b')$
$a = L$	$\omega(L, L)$	$\omega(L, H)$
	$a' = L$	$a' = H$

$$Y_{t+1}(a, a', \nu) = \left(\omega(b, b') \int_0^{s^*+\nu} \left(b' \lambda (1-\delta) k_t(s_t)\right)^{\frac{\eta-1}{\eta}} dF(s_t \mid b) + \text{no investment firms} \right)^{\frac{\eta}{\eta-1}}$$

- Higher Y_{t+1} shifts the cutoff in *every* group:

$$\begin{aligned} \nu^{(1)}(a, a') &:= \Delta s_t^*(a, a', \nu) \\ &= s^*(a, a') + \frac{\log(Y_{t+1}(\nu)/Y)}{\log \lambda} - \left(s^*(a, a') + \frac{\log(\overset{=Y}{Y_{t+1}(0)}/Y)}{\log \lambda} \right) \end{aligned}$$

- aggregate ν-**induced fraction**:

$$\nu^{(1)} := \sum_{(a,a')} \omega(a, a')\, \nu^{(1)}(a, a').$$

- The first ν of investing firms raises Y, which induces $\nu^{(1)}$ more, which raises Y again ... a GE multiplier that iterates to a new equilibrium.

Total induced investment $\gg \nu$

$$\text{Total induced investment} = \sum_{i=1}^{\infty} \text{induced fraction}^{(i)} \times \text{average induced investment}^{(i)}$$

- Scale of induced fraction

$$\lim_{\nu \to 0} \frac{\nu^{(1)}}{\omega(b, b')\,\nu} = 1$$

  - the $\nu^{(1)}$ has the *same scale* as the original fraction
    $\Rightarrow$ each $\omega(a, a')\nu^{(1)}(a, a')$ will induce the *same scale* $\nu^{(2)}(\nu^{(1)}(a, a'))$ again
    $\Rightarrow$ induced fraction will not decay through iteration.

- Scale of average induced investment

$$\left.\frac{\partial \Delta x(a, a', \nu)}{\partial \nu}\right|_{\nu=0} = (\lambda - 1)(1 - \delta)\lambda^{s^*} a^{\eta-1} \Psi Y$$

  - ν-induced average investment, $\Delta x(a, a', \nu)$, is defined as
    $\Delta x_t(a, a', \nu) := \int_{s^*}^{s^*+\nu} (\lambda - 1)(1 - \delta)\, \underbrace{\lambda^{s_t} a^{\eta-1} \Phi Y}_{k_t(s_t)}\, dF(s_t \mid a_t = a)$
    $\Rightarrow$ the derivative says $\Delta x(a, a', \nu) = O(\nu)$
    $\Rightarrow$ average induced investment will not decay faster than ν.

The estimated DSGE model

Based on Smets–Wouters (2007) and Herbst–Schorfheide (2015), augmented with the investment demand shock (two-step equilibrium).

- **Nominal rigidities:** Calvo price setting (no indexation); reduced-form **sticky wage**—partial adjustment of the wage toward the MRS:

$$w_t = \left(L_t^{\sigma_\ell}/C_t^{-\sigma}\right)^{\theta_w} w^{1-\theta_w}.$$

- **Real frictions: investment adjustment cost** $S(I_t/I_{t-1})$ and **endogenous capital utilization** u_t. Capital law of motion carries the IST shock:

$$K_{t+1} = (1-\delta)K_t + \mu_{s,t}[1 - S(I_t/I_{t-1})]I_t.$$

- **Policy:** Taylor rule (inflation + output gap + smoothing); output-stabilizing government spending.
- **Five structural shocks:** technology ε_a, IST ε_s, monetary ε_R, government ε_g, and investment demand ε_d.

Estimation: method and data

Method

- Bayesian estimation; methodology follows Smets–Wouters (2007) and Justiniano et al. (2010).
- Posterior via Metropolis–Hastings: $100{,}000$ draws, 0.5 burn-in, relatively flat priors.
- **ID vs. IST identification** rests on their different *comovement* signatures—there is no direct data series for the ID shock.

Data

- Six quarterly U.S. observables (FRED), **1983:Q1–2002:Q4**:
  - output growth, CPI inflation, fed funds rate,
  - hours, investment growth, gov.-spending growth.
- Sample fits a constant linear trend, is comparable to JPT, and avoids the ZLB.
- Measurement error (4% of each series' variance) avoids stochastic singularity.

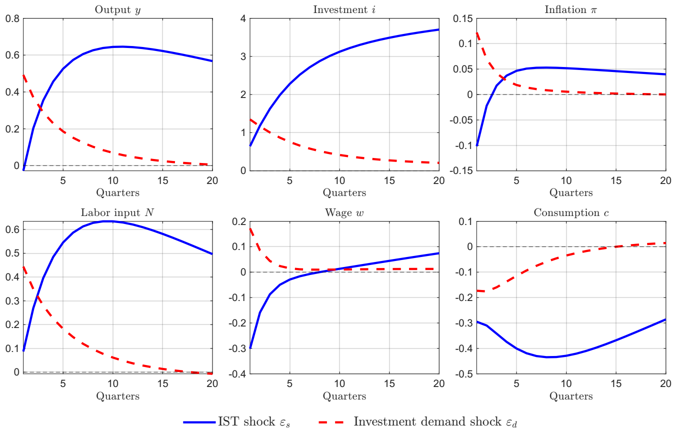
Baseline result: variance decomposition

- **ID shock** alone explains $\approx \mathbf{42}\%$ of output and $\approx \mathbf{51}\%$ of investment FEVD—a competitive driver of the cycle.
- Jointly, IST + ID account for $\approx 92\%$ of hours and $\approx 47\%$ of inflation.
- ID adds little to *hours* and is less persistent than IST ($\rho_s \approx 1$ vs. $\rho_d = 0.87$; IST is also hump-shaped through the adjustment cost).

Observable	ε_g	ε_s	ε_R	ε_a	ε_d
Output	4.63	15.55	11.00	27.12	**41.69**
Inflation	0.32	39.26	12.87	40.14	7.41
Interest rate	0.37	55.06	5.48	31.22	7.87
Hours	4.34	90.19	0.08	3.56	1.83
Investment	0.11	36.23	0.15	12.04	**51.48**
Gov. spending	99.87	0.02	0.01	0.04	0.06

Unconditional FEVD (%). ε_g gov., ε_s IST, ε_R monetary, ε_a technology, ε_d ID.

Baseline result: impulse responses



IST shock ⇒ deflationary

- Wage ↓ while hours ↑, consumption ↓, output ≈ flat.
- Strong *intertemporal substitution*: cheap investment ⇒ save more today (labor supply shifts out).

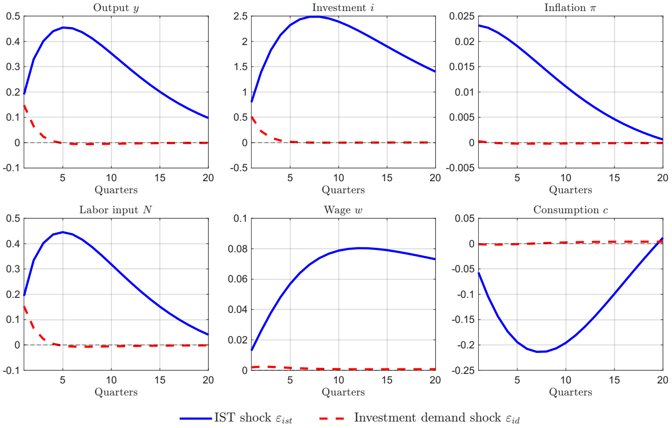
ID shock ⇒ inflationary

- Wage and hours rise *together* ⇒ labor *demand* shift dominates; output rises strongly.
- *Wealth effect*: higher future output raises lifetime wealth, so little substitution—hence comovement.

Robustness: Smets–Wouters-style model

Add habit formation, a monopolistic labor market (Erceg wage Phillips curve), and preference / price- / wage-markup shocks; add consumption and wage growth as observables.

- **Better hours fit:** $\mathrm{Var}(\hat{N}_t)$ falls $32.2 \to 9.8$ (data 6.3)—the **wage-markup shock** does the work.
- **ID shock turns marginal:** only 4.4% of output and 13.6% of investment FEVD.
- **IRFs flip:** IST is now *inflationary* with a *positive* wage response; the ID consumption response is far smoother ($\hat{c}_1$ is 0.2% of $\hat{\imath}_1$, vs. 12% in the baseline).



Conclusion

- Revisits Justiniano et al. (2010) by testing Nirei's **investment demand shock** in an estimated DSGE with *both* IST and ID shocks.
- **Baseline:** the ID shock alone explains $\approx 40\%$ of output and $\approx 50\%$ of investment FEVD—a competitive driver—and generates an *inflationary* expansion with comovement (vs. deflationary IST).
- **Limitations / fragility:**
  - no direct data identifies the ID shock—identification leans on comovement differences, hard to test over the whole parameter space;
  - imperfect fit and some extreme posteriors in the baseline;
  - results are specification-sensitive: the SW-style model shrinks the ID shock, and an i.i.d. ID shock weakens it.
- **Takeaway:** structural estimation gives genuine insight into investment-side shocks, but the model specification must be chosen carefully.